

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $y^2 - 25 =$

(2) Factor: $y^2 + 6y + 8 =$

(3) Factor: $24x^2 + 6x =$

(4) Factor: $8y^2 + 16y =$

(5) Factor: $x^2 - 36 =$

(6) Factor: $x^2 - 16 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $12x^2 + 6x =$

(2) Factor: $y^2 - 16 =$

(3) Factor: $x^2 + 6x + 8 =$

(4) Factor: $x^2 + 6x + 5 =$

(5) Factor: $9y^2 + 9y =$

(6) Factor: $x^2 + 8x + 15 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $x^2 - 36 =$

(2) Factor: $12y^2 + 16y =$

(3) Factor: $x^2 + 7x + 10 =$

(4) Factor: $x^2 + 8x + 12 =$

(5) Factor: $y^2 + 2y + 1 =$

(6) Factor: $21x^2 + 18x =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $x^2 - 9 =$

(2) Factor: $6x^2 + 10x =$

(3) Factor: $y^2 - 81 =$

(4) Factor: $x^2 + 2x + 1 =$

(5) Factor: $y^2 - 81 =$

(6) Factor: $y^2 + 9y + 20 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $x^2 + 9x + 20 =$

(2) Factor: $x^2 + 10x + 24 =$

(3) Factor: $y^2 + 7y + 12 =$

(4) Factor: $8y^2 + 10y =$

(5) Factor: $x^2 + 6x + 8 =$

(6) Factor: $y^2 - 81 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $x^2 - 16 =$

(2) Factor: $y^2 - 25 =$

(3) Factor: $18y^2 + 15y =$

(4) Factor: $y^2 - 4 =$

(5) Factor: $12y^2 + 10y =$

(6) Factor: $6y^2 + 6y =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $y^2 - 36 =$

(2) Factor: $x^2 + 5x + 4 =$

(3) Factor: $21y^2 + 12y =$

(4) Factor: $x^2 - 64 =$

(5) Factor: $12x^2 + 18x =$

(6) Factor: $24y^2 + 20y =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $18y^2 + 9y =$

(2) Factor: $y^2 + 7y + 12 =$

(3) Factor: $12y^2 + 12y =$

(4) Factor: $x^2 - 81 =$

(5) Factor: $10y^2 + 6y =$

(6) Factor: $16x^2 + 20x =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $x^2 - 16 =$

(2) Factor: $16y^2 + 16y =$

(3) Factor: $x^2 - 25 =$

(4) Factor: $y^2 + 2y + 1 =$

(5) Factor: $21x^2 + 12x =$

(6) Factor: $y^2 + 7y + 10 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $y^2 - 25 =$

(2) Factor: $x^2 + 6x + 8 =$

(3) Factor: $y^2 + 8y + 12 =$

(4) Factor: $12x^2 + 4x =$

(5) Factor: $8y^2 + 10y =$

(6) Factor: $x^2 - 64 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $y^2 + 7y + 6 =$

(2) Factor: $24x^2 + 16x =$

(3) Factor: $y^2 - 49 =$

(4) Factor: $24y^2 + 16y =$

(5) Factor: $y^2 - 81 =$

(6) Factor: $12x^2 + 18x =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $x^2 + 8x + 15 =$

(2) Factor: $x^2 - 64 =$

(3) Factor: $y^2 + 8y + 15 =$

(4) Factor: $y^2 + 7y + 12 =$

(5) Factor: $12y^2 + 6y =$

(6) Factor: $y^2 + 6y + 5 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $32y^2 + 16y =$

(2) Factor: $12y^2 + 15y =$

(3) Factor: $y^2 + 4y + 3 =$

(4) Factor: $y^2 + 7y + 12 =$

(5) Factor: $8y^2 + 16y =$

(6) Factor: $20y^2 + 16y =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $21y^2 + 15y =$

(2) Factor: $x^2 + 6x + 9 =$

(3) Factor: $x^2 - 9 =$

(4) Factor: $x^2 + 3x + 2 =$

(5) Factor: $x^2 - 4 =$

(6) Factor: $y^2 - 16 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $y^2 + 2y + 1 =$

(2) Factor: $x^2 - 49 =$

(3) Factor: $y^2 - 81 =$

(4) Factor: $8y^2 + 6y =$

(5) Factor: $x^2 - 81 =$

(6) Factor: $y^2 + 3y + 2 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $x^2 - 25 =$

(2) Factor: $y^2 + 5y + 6 =$

(3) Factor: $12x^2 + 8x =$

(4) Factor: $x^2 + 3x + 2 =$

(5) Factor: $8y^2 + 10y =$

(6) Factor: $x^2 + 6x + 5 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $y^2 + 8y + 15 =$

(2) Factor: $x^2 + 7x + 10 =$

(3) Factor: $y^2 + 9y + 20 =$

(4) Factor: $x^2 - 64 =$

(5) Factor: $x^2 + 4x + 4 =$

(6) Factor: $y^2 - 25 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $x^2 - 36 =$

(2) Factor: $x^2 - 25 =$

(3) Factor: $12y^2 + 20y =$

(4) Factor: $y^2 + 10y + 24 =$

(5) Factor: $y^2 + 4y + 4 =$

(6) Factor: $x^2 + 9x + 20 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $y^2 + 6y + 5 =$

(2) Factor: $32x^2 + 24x =$

(3) Factor: $12x^2 + 16x =$

(4) Factor: $x^2 - 25 =$

(5) Factor: $y^2 + 6y + 5 =$

(6) Factor: $x^2 - 81 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $15y^2 + 9y =$

(2) Factor: $y^2 - 36 =$

(3) Factor: $x^2 + 9x + 18 =$

(4) Factor: $x^2 - 4 =$

(5) Factor: $x^2 + 7x + 12 =$

(6) Factor: $28y^2 + 24y =$

Section 4 - Solutions

31a-1: $(y + 5)(y - 5)$

31a-2: $(y + 4)(y + 2)$

31a-3: $3x(8x + 2)$

31a-4: $4y(2y + 4)$

31a-5: $(x + 6)(x - 6)$

31a-6: $(x + 4)(x - 4)$

31b-1: $(x + 3)(x - 3)$

31b-2: $2x(3x + 5)$

31b-3: $(y + 9)(y - 9)$

31b-4: $(x + 1)(x + 1)$

31b-5: $(y + 9)(y - 9)$

31b-6: $(y + 4)(y + 5)$

32a-1: $3x(4x + 2)$

32a-2: $(y + 4)(y - 4)$

32a-3: $(x + 2)(x + 4)$

Section 4 - Solutions (continued)

33b-1: $3y(6y + 3)$

33b-2: $(y + 3)(y + 4)$

33b-3: $2y(6y + 6)$

33b-4: $(x + 9)(x - 9)$

33b-5: $2y(5y + 3)$

33b-6: $4x(4x + 5)$

34a-1: $(x + 4)(x - 4)$

34a-2: $(y + 5)(y - 5)$

34a-3: $3y(6y + 5)$

34a-4: $(y + 2)(y - 2)$

34a-5: $2y(6y + 5)$

34a-6: $3y(2y + 2)$

34b-1: $(y + 6)(y - 6)$

34b-2: $(x + 1)(x + 4)$

34b-3: $3y(7y + 4)$

Section 4 - Solutions (continued)

34b-4: $(x + 8)(x - 8)$

34b-5: $3x(4x + 6)$

34b-6: $4y(6y + 5)$

35a-1: $(x + 4)(x - 4)$

35a-2: $4y(4y + 4)$

35a-3: $(x + 5)(x - 5)$

35a-4: $(y + 1)(y + 1)$

35a-5: $3x(7x + 4)$

35a-6: $(y + 2)(y + 5)$

35b-1: $(x + 3)(x + 5)$

35b-2: $(x + 8)(x - 8)$

35b-3: $(y + 3)(y + 5)$

35b-4: $(y + 3)(y + 4)$

35b-5: $3y(4y + 2)$

35b-6: $(y + 1)(y + 5)$

Section 4 - Solutions (continued)

32a-4: $(x + 1)(x + 5)$

32a-5: $3y(3y + 3)$

32a-6: $(x + 3)(x + 5)$

32b-1: $(x + 6)(x - 6)$

32b-2: $4y(3y + 4)$

32b-3: $(x + 2)(x + 5)$

32b-4: $(x + 2)(x + 6)$

32b-5: $(y + 1)(y + 1)$

32b-6: $3x(7x + 6)$

33a-1: $(x + 4)(x + 5)$

33a-2: $(x + 4)(x + 6)$

33a-3: $(y + 3)(y + 4)$

33a-4: $2y(4y + 5)$

33a-5: $(x + 2)(x + 4)$

33a-6: $(y + 9)(y - 9)$

Section 4 - Solutions (continued)

36a-1: $(y + 5)(y - 5)$

36a-2: $(x + 4)(x + 2)$

36a-3: $(y + 2)(y + 6)$

36a-4: $2x(6x + 2)$

36a-5: $2y(4y + 5)$

36a-6: $(x + 8)(x - 8)$

36b-1: $(y + 1)(y + 6)$

36b-2: $4x(6x + 4)$

36b-3: $(y + 7)(y - 7)$

36b-4: $4y(6y + 4)$

36b-5: $(y + 9)(y - 9)$

36b-6: $3x(4x + 6)$

37a-1: $4y(8y + 4)$

37a-2: $3y(4y + 5)$

37a-3: $(y + 3)(y + 1)$

Section 4 - Solutions (continued)

38b-1: $(y + 1)(y + 1)$

38b-2: $(x + 7)(x - 7)$

38b-3: $(y + 9)(y - 9)$

38b-4: $2y(4y + 3)$

38b-5: $(x + 9)(x - 9)$

38b-6: $(y + 2)(y + 1)$

39a-1: $(y + 3)(y + 5)$

39a-2: $(x + 2)(x + 5)$

39a-3: $(y + 4)(y + 5)$

39a-4: $(x + 8)(x - 8)$

39a-5: $(x + 2)(x + 2)$

39a-6: $(y + 5)(y - 5)$

39b-1: $3y(5y + 3)$

39b-2: $(y + 6)(y - 6)$

39b-3: $(x + 3)(x + 6)$

Section 4 - Solutions (continued)

39b-4: $(x + 2)(x - 2)$

39b-5: $(x + 4)(x + 3)$

39b-6: $4y(7y + 6)$

40a-1: $(x + 6)(x - 6)$

40a-2: $(x + 5)(x - 5)$

40a-3: $4y(3y + 5)$

40a-4: $(y + 4)(y + 6)$

40a-5: $(y + 2)(y + 2)$

40a-6: $(x + 4)(x + 5)$

40b-1: $(y + 1)(y + 5)$

40b-2: $4x(8x + 6)$

40b-3: $4x(3x + 4)$

40b-4: $(x + 5)(x - 5)$

40b-5: $(y + 1)(y + 5)$

40b-6: $(x + 9)(x - 9)$

Section 4 - Solutions (continued)

37a-4: $(y + 4)(y + 3)$

37a-5: $4y(2y + 4)$

37a-6: $4y(5y + 4)$

37b-1: $(x + 5)(x - 5)$

37b-2: $(y + 3)(y + 2)$

37b-3: $4x(3x + 2)$

37b-4: $(x + 2)(x + 1)$

37b-5: $2y(4y + 5)$

37b-6: $(x + 1)(x + 5)$

38a-1: $3y(7y + 5)$

38a-2: $(x + 3)(x + 3)$

38a-3: $(x + 3)(x - 3)$

38a-4: $(x + 1)(x + 2)$

38a-5: $(x + 2)(x - 2)$

38a-6: $(y + 4)(y - 4)$